Week 9: Sales and Customer Analytics System

The e-commerce company wants to analyze its sales data to understand customer purchasing behavior, product performance, and revenue trends. The management team needs reports to identify top customers, best-selling products, and profitable categories.

1. Apply SQL Aggregate Functions

Aggregate functions perform calculations on multiple records and return a single summarized value.

Functions Covered: Function	**Purpose**	**Example**
COUNT()	Counts number of records	Total customers/orders
SUM()	Calculates total value	Total sales amount
AVG()	Finds average value	Average order value
MIN()	Finds minimum value	Lowest product price
MAX()	Finds maximum value	Highest produ

2. Generate Total Sales

Reports Tasks:

Write SQL queries to calculate:

1. Total number of orders placed.

```
SQL> SELECT COUNT(*) AS Number_Of_Orders_Placed FROM Orders;

NUMBER_OF_ORDERS_PLACED
-----------------------
                     14
```

2. Total revenue generated from sales.

```
SQL> SELECT SUM(TotalAmount) AS Total_Revenue FROM Orders;

TOTAL_REVENUE
-------------
       206200
```

3. Average order value.

```
SQL> SELECT ROUND(AVG(TotalAmount),2) AS Average_Order_Value FROM Orders;

AVERAGE_ORDER_VALUE
-------------------
          14728.57
```

4. Highest and lowest order amount.

```
SQL> SELECT MAX(TotalAmount) AS Highest_Order,
  2  MIN(TotalAmount) AS Lowest_Order
  3  FROM Orders;

HIGHEST_ORDER LOWEST_ORDER
------------- ------------
        55850         2450
```

5. Total sales for a specific period.

```
SQL> SELECT SUM(TotalAmount) AS Total_Revenue FROM Orders
  2  WHERE OrderDate BETWEEN DATE '2026-06-01' AND DATE '2026-08-31';

TOTAL_REVENUE
-------------
        37450
```

3. Customer Purchase Analysis

1. Total number of orders placed by each customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS TOTAL_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_I CUSTOMER_NAME                          TOTAL_ORDERS
---------- ---------------------------------------- ------------
CUST000001 Arjun Kumar                                       1
CUST000015 Nithin Murugan                                    1
CUST000003 Rahul Menon                                       1
CUST000004 Ananya Iyer                                       1
CUST000005 Vikram Reddy                                      1
CUST000006 Meera Krishnan                                    1
CUST000007 Suresh Nair                                       1
CUST000008 Kavya Rajan                                       1
CUST000009 Karthik Subramanian                               1
CUST000010 Divya Pillai                                      1
CUST000011 Aditya Verma                                      1

CUSTOMER_I CUSTOMER_NAME                          TOTAL_ORDERS
---------- ---------------------------------------- ------------
CUST000012 Sneha Patel                                       1
CUST000002 Priya Sharma                                      1
CUST000014 Lakshmi Venkatesan                                1
CUST000013 Rohan Bose                                        0

15 rows selected.
```

2. Total amount spent by each customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      NVL(SUM(O.TOTALAMOUNT), 0) AS TOTAL_AMOUNT_SPENT
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUSTOMER_I CUSTOMER_NAME                          TOTAL_AMOUNT_SPENT
---------- ---------------------------------------- ------------------
CUST000001 Arjun Kumar                                        55850
CUST000014 Lakshmi Venkatesan                                 28000
CUST000003 Rahul Menon                                        28000
CUST000005 Vikram Reddy                                       22000
CUST000009 Karthik Subramanian                                18500
CUST000015 Nithin Murugan                                      8350
CUST000004 Ananya Iyer                                         8300
CUST000011 Aditya Verma                                        7000
CUST000006 Meera Krishnan                                      7000
CUST000007 Suresh Nair                                         6500
CUST000008 Kavya Rajan                                         5700

CUSTOMER_I CUSTOMER_NAME                          TOTAL_AMOUNT_SPENT
---------- ---------------------------------------- ------------------
CUST000010 Divya Pillai                                        5200
CUST000002 Priya Sharma                                        3350
CUST000012 Sneha Patel                                         2450
CUST000013 Rohan Bose                                             0

15 rows selected.
```

NVL() function replaces NULL with a specified value. Here for the customer with no purchase the Amount Spent will be zero.

3. Average spending per customer.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      ROUND(NVL(AVG(O.TOTALAMOUNT), 0), 2) AS AVERAGE_SPENDING
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY AVERAGE_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME                          AVERAGE_SPENDING
---------- ------------------------------------- ----------------
CUST000001 Arjun Kumar                                      55850
CUST000014 Lakshmi Venkatesan                               28000
CUST000003 Rahul Menon                                      28000
CUST000005 Vikram Reddy                                     22000
CUST000009 Karthik Subramanian                              18500
CUST000015 Nithin Murugan                                    8350
CUST000004 Ananya Iyer                                       8300
CUST000011 Aditya Verma                                      7000
CUST000006 Meera Krishnan                                    7000
CUST000007 Suresh Nair                                       6500
CUST000008 Kavya Rajan                                       5700

CUSTOMER_I CUSTOMER_NAME                          AVERAGE_SPENDING
---------- ------------------------------------- ----------------
CUST000010 Divya Pillai                                      5200
CUST000002 Priya Sharma                                      3350
CUST000012 Sneha Patel                                       2450
CUST000013 Rohan Bose                                           0

15 rows selected.
```

4. Customers with maximum purchase amount.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_PURCHASE_AMOUNT
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING SUM(O.TOTALAMOUNT) = (
 13      SELECT MAX(TOTAL_SPENDING)
 14      FROM (
 15          SELECT SUM(TOTALAMOUNT) AS TOTAL_SPENDING
 16          FROM ORDERS
 17          GROUP BY CUSTOMERID
 18      )
 19  );

CUSTOMER_I CUSTOMER_NAME                          TOTAL_PURCHASE_AMOUNT
---------- ------------------------------------- ---------------------
CUST000001 Arjun Kumar                                           55850
```

5. Customers with fewer purchases.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  LEFT JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) <= 1
 13  ORDER BY NUMBER_OF_ORDERS;

CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ---------------------------------------- ----------------
CUST000013 Rohan Bose                                            0
CUST000015 Nithin Murugan                                        1
CUST000003 Rahul Menon                                           1
CUST000004 Ananya Iyer                                           1
CUST000005 Vikram Reddy                                          1
CUST000006 Meera Krishnan                                        1
CUST000007 Suresh Nair                                           1
CUST000008 Kavya Rajan                                           1
CUST000009 Karthik Subramanian                                   1
CUST000010 Divya Pillai                                          1
CUST000011 Aditya Verma                                          1

CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ---------------------------------------- ----------------
CUST000012 Sneha Patel                                           1
CUST000014 Lakshmi Venkatesan                                    1
CUST000001 Arjun Kumar                                           1
CUST000002 Priya Sharma                                          1

15 rows selected.
```

4. Find Top Customers Based on Purchase Amount

1. Top 5 customers based on total spending.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  ORDER BY TOTAL_SPENDING DESC
 13  FETCH FIRST 5 ROWS ONLY;

CUSTOMER_I CUSTOMER_NAME                          TOTAL_SPENDING
---------- ---------------------------------------- --------------
CUST000001 Arjun Kumar                                      55850
CUST000003 Rahul Menon                                      28000
CUST000014 Lakshmi Venkatesan                               28000
CUST000005 Vikram Reddy                                     22000
CUST000009 Karthik Subramanian                              18500
```

2. Customers with maximum number of orders.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) = (
 13      SELECT MAX(ORDER_COUNT)
 14      FROM (
 15          SELECT COUNT(*) AS ORDER_COUNT
 16          FROM ORDERS
 17          GROUP BY CUSTOMERID
 18      )
 19  )
 20  ORDER BY CUSTOMER_NAME;

CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ------------------------------------ ---------------
CUST000011 Aditya Verma                                         1
CUST000004 Ananya Iyer                                          1
CUST000001 Arjun Kumar                                          1
CUST000010 Divya Pillai                                         1
CUST000009 Karthik Subramanian                                  1
CUST000008 Kavya Rajan                                          1
CUST000014 Lakshmi Venkatesan                                   1
CUST000006 Meera Krishnan                                       1
CUST000015 Nithin Murugan                                       1
CUST000002 Priya Sharma                                         1
CUST000003 Rahul Menon                                          1

CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ------------------------------------ ---------------
CUST000012 Sneha Patel                                          1
CUST000007 Suresh Nair                                          1
CUST000005 Vikram Reddy                                         1

14 rows selected.
```

3. Frequent customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7      ON C.CUSTOMER_ID = O.CUSTOMERID
  8  GROUP BY
  9      C.CUSTOMER_ID,
 10      C.FIRST_NAME,
 11      C.LAST_NAME
 12  HAVING COUNT(O.ORDERID) >= 2
 13  ORDER BY NUMBER_OF_ORDERS DESC;

no rows selected
```

4. High-value customers.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS,
  5      SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMERID
  9  GROUP BY
 10      C.CUSTOMER_ID,
 11      C.FIRST_NAME,
 12      C.LAST_NAME
 13  HAVING SUM(O.TOTALAMOUNT) > 20000
 14  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ---------------------------------------- ---------------
TOTAL_SPENDING
--------------
CUST000001 Arjun Kumar                                           1
         55850

CUST000014 Lakshmi Venkatesan                                    1
         28000

CUST000003 Rahul Menon                                           1
         28000


CUSTOMER_I CUSTOMER_NAME                          NUMBER_OF_ORDERS
---------- ---------------------------------------- ---------------
TOTAL_SPENDING
--------------
CUST000005 Vikram Reddy                                          1
         22000
```

5. Identify Best-Selling Products

1. Products with maximum sales quantity.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT PRODUCT_NAME                                  TOTAL_QUANTITY_SOLD
------- --------------------------------------------- -------------------
P000006 USB Cable                                                       6
P000005 Smartphone                                                      6
P000015 Wireless Earbuds                                                3
P000008 Smart Watch                                                     2
P000012 External Hard Disk                                              2
P000009 Headphones                                                      2
P000001 Laptop                                                          1
P000013 Gaming Monitor                                                  1
P000010 Tablet                                                          1
P000014 Laptop Stand                                                    1
P000002 Wireless Mouse                                                  1

PRODUCT PRODUCT_NAME                                  TOTAL_QUANTITY_SOLD
------- --------------------------------------------- -------------------
P000003 Mechanical Keyboard                                             1
P000004 Bluetooth Speaker                                               1

13 rows selected.
```

2. Products generating highest revenue.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  5      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  6  FROM PRODUCT P
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      P.PRODUCT_ID,
 11      P.PRODUCT_NAME
 12  ORDER BY TOTAL_REVENUE DESC;

PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000005 Smartphone                                                     6
       168000

P000001 Laptop                                                         1
        55000

P000010 Tablet                                                         1
        22000
```

```
PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000013 Gaming Monitor                                                 1
        18500

P000008 Smart Watch                                                    2
        13000

P000015 Wireless Earbuds                                               3
        10500


PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000012 External Hard Disk                                             2
        10400

P000009 Headphones                                                     2
         8400

P000004 Bluetooth Speaker                                              1
         3200


PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000006 USB Cable                                                      6
         2700

P000003 Mechanical Keyboard                                            1
         2500

P000014 Laptop Stand                                                   1
         1500


PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
TOTAL_REVENUE
-------------
P000002 Wireless Mouse                                                 1
          850


13 rows selected.
```

3. Least-selling products.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  5  FROM PRODUCT P
  6  LEFT JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY QUANTITY_SOLD ASC;

PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
P000007 Power Bank                                                     0
P000011 Web Camera                                                     0
P000004 Bluetooth Speaker                                              1
P000003 Mechanical Keyboard                                            1
P000001 Laptop                                                         1
P000002 Wireless Mouse                                                 1
P000014 Laptop Stand                                                   1
P000013 Gaming Monitor                                                 1
P000010 Tablet                                                         1
P000012 External Hard Disk                                             2
P000009 Headphones                                                     2

PRODUCT PRODUCT_NAME                                       QUANTITY_SOLD
------- -------------------------------------------------- -------------
P000008 Smart Watch                                                    2
P000015 Wireless Earbuds                                               3
P000005 Smartphone                                                     6
P000006 USB Cable                                                      6

15 rows selected
```

4. Products requiring promotion.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      P.PRICE,
  5      P.STOCK_QUANTITY,
  6      NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD
  7  FROM PRODUCT P
  8  LEFT JOIN ORDERDETAIL OD
  9      ON P.PRODUCT_ID = OD.PRODUCTID
 10  GROUP BY
 11      P.PRODUCT_ID,
 12      P.PRODUCT_NAME,
 13      P.PRICE,
 14      P.STOCK_QUANTITY
 15  HAVING NVL(SUM(OD.QUANTITY), 0) <= 1
 16     AND P.STOCK_QUANTITY > 0
 17  ORDER BY QUANTITY_SOLD, P.PRODUCT_NAME;

PRODUCT PRODUCT_NAME                                         PRICE
------- -------------------------------------------------- ----------
STOCK_QUANTITY QUANTITY_SOLD
-------------- -------------
P000007 Power Bank                                           1800
            75             0

P000011 Web Camera                                           2800
            50             0

P000004 Bluetooth Speaker                                    3200
            45             1


PRODUCT PRODUCT_NAME                                         PRICE
------- -------------------------------------------------- ----------
STOCK_QUANTITY QUANTITY_SOLD
-------------- -------------
P000013 Gaming Monitor                                      18500
            20             1

P000001 Laptop                                              55000
            25             1

P000014 Laptop Stand                                         1500
            80             1


PRODUCT PRODUCT_NAME                                         PRICE
------- -------------------------------------------------- ----------
STOCK_QUANTITY QUANTITY_SOLD
-------------- -------------
P000003 Mechanical Keyboard                                  2500
            60             1

P000010 Tablet                                              22000
            30             1

P000002 Wireless Mouse                                        850
           100             1


9 rows selected.
```

6. Perform Category-wise Sales Analysis

1. Total sales generated by each category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY TOTAL_SALES DESC;

CATEGORY_NAME                  TOTAL_SALES
------------------------------ -----------
Mobile Devices                      190000
Laptops                              55000
Audio Devices                        22100
Monitors                             18500
Wearables                            13000
Storage Devices                      10400
Computer Accessories                  3350
Power Accessories                     2700
Laptop Accessories                    1500

9 rows selected.
```

2. Number of products sold in each category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY) AS PRODUCTS_SOLD
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY PRODUCTS_SOLD DESC;

CATEGORY_NAME                  PRODUCTS_SOLD
------------------------------ -------------
Mobile Devices                             7
Power Accessories                          6
Audio Devices                              6
Computer Accessories                       2
Wearables                                  2
Storage Devices                            2
Laptop Accessories                         1
Laptops                                    1
Monitors                                   1

9 rows selected.
```

3. Highest revenue-generating category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  HAVING SUM(OD.QUANTITY * OD.PRICE) = (
 13      SELECT MAX(CATEGORY_REVENUE)
 14      FROM (
 15          SELECT SUM(OD2.QUANTITY * OD2.PRICE) AS CATEGORY_REVENUE
 16          FROM CATEGORY C2
 17          JOIN PRODUCT P2
 18              ON C2.CATEGORY_ID = P2.CATEGORY_ID
 19          JOIN ORDERDETAIL OD2
 20              ON P2.PRODUCT_ID = OD2.PRODUCTID
 21          GROUP BY C2.CATEGORY_ID
 22      )
 23  );

CATEGORY_NAME                  TOTAL_REVENUE
------------------------------ -------------
Mobile Devices                        190000
```

4. Average sales per category.

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      ROUND(AVG(OD.QUANTITY * OD.PRICE), 2) AS AVERAGE_SALES
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6      ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDERDETAIL OD
  8      ON P.PRODUCT_ID = OD.PRODUCTID
  9  GROUP BY
 10      C.CATEGORY_ID,
 11      C.CATEGORY_NAME
 12  ORDER BY AVERAGE_SALES DESC;

CATEGORY_NAME                  AVERAGE_SALES
------------------------------ -------------
Mobile Devices                         95000
Laptops                                55000
Monitors                               18500
Wearables                               6500
Storage Devices                         5200
Audio Devices                        3683.33
Computer Accessories                    1675
Laptop Accessories                      1500
Power Accessories                        900

9 rows selected.
```

7. Generate Business Analytics Reports Report 1: Sales Performance Report

- Total sales

- Number of orders
- Average order value

Highest order value

```
SQL> SELECT
  2      SUM(TOTALAMOUNT) AS TOTAL_SALES,
  3      COUNT(ORDERID) AS NUMBER_OF_ORDERS,
  4      ROUND(AVG(TOTALAMOUNT), 2) AS AVERAGE_ORDER_VALUE,
  5      MAX(TOTALAMOUNT) AS HIGHEST_ORDER_VALUE
  6  FROM ORDERS;

TOTAL_SALES NUMBER_OF_ORDERS AVERAGE_ORDER_VALUE HIGHEST_ORDER_VALUE
----------- ---------------- ------------------- -------------------
     206200               14            14728.57               55850
```

Report 2: Customer Analytics Report

- Customer name
- Number of orders
- Total spending
- Purchase frequency

```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      COUNT(O.ORDERID) AS NUMBER_OF_ORDERS,
  4      NVL(SUM(O.TOTALAMOUNT), 0) AS TOTAL_SPENDING,
  5      COUNT(O.ORDERID) AS PURCHASE_FREQUENCY
  6  FROM CUSTOMER C
  7  LEFT JOIN ORDERS O
  8      ON C.CUSTOMER_ID = O.CUSTOMERID
  9  GROUP BY
 10      C.CUSTOMER_ID,
 11      C.FIRST_NAME,
 12      C.LAST_NAME
 13  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Arjun Kumar                                           1          55850
                 1

Lakshmi Venkatesan                                    1          28000
                 1

Rahul Menon                                           1          28000
                 1


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Vikram Reddy                                          1          22000
                 1

Karthik Subramanian                                   1          18500
                 1

Nithin Murugan                                        1           8350
                 1
```

```
CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Ananya Iyer                                           1           8300
                 1

Aditya Verma                                          1           7000
                 1

Meera Krishnan                                        1           7000
                 1


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Suresh Nair                                           1           6500
                 1

Kavya Rajan                                           1           5700
                 1

Divya Pillai                                          1           5200
                 1


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
------------------
Priya Sharma                                          1           3350
                 1

Sneha Patel                                           1           2450
                 1

Rohan Bose                                            0              0
                 0


15 rows selected.
```

Report 3: Product Performance Report

- Product name
- Quantity sold
- Revenue generated

```
SQL> SELECT
  2      P.PRODUCT_NAME,
  3      SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS REVENUE_GENERATED
  5  FROM PRODUCT P
  6  JOIN ORDERDETAIL OD
  7      ON P.PRODUCT_ID = OD.PRODUCTID
  8  GROUP BY
  9      P.PRODUCT_ID,
 10      P.PRODUCT_NAME
 11  ORDER BY REVENUE_GENERATED DESC;
```

```
PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Smartphone                                                     6
          168000

Laptop                                                         1
           55000

Tablet                                                         1
           22000


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Gaming Monitor                                                 1
           18500

Smart Watch                                                    2
           13000

Wireless Earbuds                                               3
           10500


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
External Hard Disk                                             2
           10400

Headphones                                                     2
            8400

Bluetooth Speaker                                              1
            3200


PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
USB Cable                                                      6
            2700

Mechanical Keyboard                                            1
            2500

Laptop Stand                                                   1
            1500
```

```
PRODUCT_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
REVENUE_GENERATED
-----------------
Wireless Mouse                                                 1
             850


13 rows selected.
```

Report 4: Category Analysis Report

- Category name
- Total products sold
- Total revenue

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(OD.QUANTITY) AS TOTAL_PRODUCTS_SOLD,
  4      SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  5  FROM CATEGORY C
  6  JOIN PRODUCT P
  7      ON C.CATEGORY_ID = P.CATEGORY_ID
  8  JOIN ORDERDETAIL OD
  9      ON P.PRODUCT_ID = OD.PRODUCTID
 10  GROUP BY
 11      C.CATEGORY_ID,
 12      C.CATEGORY_NAME
 13  ORDER BY TOTAL_REVENUE DESC;

CATEGORY_NAME                  TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------------------ ------------------- -------------
Mobile Devices                                   7        190000
Laptops                                          1         55000
Audio Devices                                    6         22100
Monitors                                         1         18500
Wearables                                        2         13000
Storage Devices                                  2         10400
Computer Accessories                             2          3350
Power Accessories                                6          2700
Laptop Accessories                               1          1500

9 rows selected.
```